



SIMATS Online Programming Lab

JAVA PROGRAMMING



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Exception- Practice- CO3

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QUESTIONS

Q1

Exception
10 marks

**Q2**

IllegalArgumentException
10 marks

**Q3**

user-defined exception named
WeakPasswordException.
10 marks

**Q4**

Email Validation
10 marks

**Q5**

Banking Transaction Using
Exception Propagation
10 marks



5/5 answered

Q5

Banking Transaction Using Exception Propagation

10
marks

Question 5: Banking Transaction Using Exception Propagation

Problem Statement

Write a Java program to simulate a banking transaction using three methods:

withdraw()

validateBalance()

processTransaction()

If the withdrawal amount exceeds the account balance, throw an exception from processTransaction(). Allow it to propagate through the other methods and handle it only in main().

Sample Input

Balance:

5000

Withdraw:

7000

Sample Output

Exception propagated to main().

InsufficientBalanceException: Withdrawal amount exceeds available balance.

Your Code

```
1 public class BankingSystem {
2
3     public static void main(String[] args)
4     {
5         double balance = 5000;
6         double withdrawAmount = 7000;
7
8         try {
9             processTransaction(balance, withdrawAmount);
10        } catch (InsufficientBalanceException e) {
11            System.out.println("Exception: " + e.getMessage());
12            System.out.println("Insufficient balance for withdrawal.");
13        }
14
15        public static void processTransaction(double balance, double withdrawAmount)
16        {
17            withdraw(balance, withdrawAmount);
18        }
19
20        public static void withdraw(double balance, double withdrawAmount)
21        {
22            validateBalance(balance, withdrawAmount);
23            System.out.println("Withdrawal successful.");
24        }
25
26        public static void validateBalance(double balance, double withdrawAmount)
27        {
28            if (withdrawAmount > balance) {
29                throw new InsufficientBalanceException("Insufficient balance for withdrawal.");
30            }
31        }
32
33        class InsufficientBalanceException extends Exception {
34            public InsufficientBalanceException(String message) {
35                super(message);
36            }
37        }
38    }
39 }
```

Input

Balance:
5000
Withdraw:
7000

Output

```
Exception propagated to main().  
InsufficientBalanceException: Withdrawal  
amount exceeds available balance.
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3

**10.00 / 10 marks** **Submitted**